power supply – prepurchase

- https://analogti.com/products/20kv-1ma-dc-dc-high-voltage-power-supply-24v-input-ahv24v20kv1maw/
    - Relatively low-cost 20kV power supply
- In general, if we can get a proper high vacuum established, we can lower our voltage requirements to 1-3kV, but they become greedier for current instead
    - One exception is for emitter-spraying thrusters
    - good vacuum specs lower power supply costs by a significant amount

power supply – in house

- See previous notes (specifically for ionic wind capable) (20KV)
    - allows up to 25mm gap distance, can be combined with a reversed polarity identical module to be upgraded to the 40KV dual polarity system once more finding is procured, ~1-3mA current capacity, a bit of a stretch for multi-stage
    - requires more consideration of insulated experimental environment
    - If needed and not too obstructive, can significantly reduce costs by looking at the lower end of this current range
- In general, if we can get a proper high vacuum established, we can lower our voltage requirements to 1-3kV, but they become greedier for current instead
    - One exception is for electrospray thrusters
        - I would like to take a solid look at electrospray thrusters because they do seem a fair bit more budget friendly and have very interesting applications that we may be able to mess with in the lab
        - Caleb's simulations and stuff may become even more important than before
    - good vacuum specs lower power supply costs by a significant amount
- I think for power supply I lean towards in house because it's a relatively well documented process and can allow us to determine our experimental needs based on the status of the vacuum chamber and customize around that

High vacuum chamber – prepurchase

- Going to be very expensive, but have determined that we want to be looking in the 30-50L scale of chambers
- To be completely honest I believe this is not a real option >$8k very prohibative

High vacuum chamber – in house

- Very possibly viable example project https://www.instructables.com/Personal-Research-Project-Into-Ultra-High-Vacuum-P/
    - The problem remains the procurement of a turbopump for not an ungodly price
- One cheaper alternative is an oil pump, but it can contaminate the electrodes and contribute error to experimentation
    - The effects can significantly be mitigated / accommodated for through careful experimental procedure and control testing
- For either option, if EPPL wants us to pursue a proper vaccuum chamber system rather than just EAD thrusters the lab will have to help source funding for thousands of dollars of pump cost (~4k-6k)

Conclusion:

Keep our research on track for now with EAD thrusters, which are much much MUCH more budget feasible, looking on the scale of a few hundred dollars for a 20kV power supply and that is most of the cost. As such, I will continue to completely spec out such a power supply so that we can apply to grants and get it being made and simulations being conducted with EAD. While I believe the cost of a high vacuum capable pump is genuinely prohibitive for us, I would like to see if the group has any interest in the concepts of oil-based pumps or electrospray thrusters as I plan to compile a full bill of materials for at least one vacuum chamber so that we may get the opportunity to act upon it with leftover department funding, a large grant, etc.